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# Heart Disease Prediction Using Machine Learning

**Predictive Analytics**
CSE3141

Machine Learning Project Report

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# 1. Abstract

This project applies machine learning classification techniques to predict the presence of heart disease from patient-related medical attributes. The project uses a dataset containing 918 patient records and 12 columns. Data inspection, cleaning, exploratory data analysis, categorical encoding, feature scaling, model training, and model comparison were performed. Five classification algorithms were evaluated: Logistic Regression, K-Nearest Neighbors, Naive Bayes, Decision Tree, and Support Vector Machine. Based on the recorded notebook results, Logistic Regression produced the highest accuracy of 86.96% and an F1-score of 88.57%.

# 2. Introduction

Heart disease prediction can be approached as a supervised machine learning classification problem. In this project, patient attributes are used as input features and the HeartDisease column is used as the target variable. The objective is to compare different classification algorithms and identify the model that performs best on the available test data.

# 3. Problem Statement

Develop a machine learning classification system capable of predicting whether a patient has heart disease using the medical and clinical features available in the dataset.

# 4. Dataset Description

The dataset used in the notebook contains 918 records and 12 columns. The columns are Age, Sex, ChestPainType, RestingBP, Cholesterol, FastingBS, RestingECG, MaxHR, ExerciseAngina, Oldpeak, ST_Slope, and HeartDisease. The target variable HeartDisease contains two classes: 0 and 1. The notebook records 508 instances of class 1 and 410 instances of class 0. No missing values and no duplicate records were found during the initial checks.

# 5. Data Preprocessing and Cleaning

The notebook first inspected the dataset using shape, data types, descriptive statistics, duplicate checks, class counts, and missing-value checks. The dataset contained zero missing values and zero duplicate rows. During cleaning, zero values in Cholesterol and RestingBP were treated as values to be replaced. The notebook calculated the mean of the non-zero Cholesterol values as 244.6354 and used it to replace zero Cholesterol values. Similarly, the mean of the non-zero RestingBP values was calculated as 132.5409 and used to replace zero RestingBP values.

Categorical variables were converted to numerical variables using pandas get_dummies with drop_first=True. Numerical columns including Age, RestingBP, Cholesterol, MaxHR, and Oldpeak were standardized using StandardScaler. The data was then separated into features (X) and the HeartDisease target (y), followed by an 80:20 training/testing split with random_state=42.

# 6. Exploratory Data Analysis

The notebook performs exploratory analysis using distribution plots for Age, RestingBP, Cholesterol, and MaxHR. It also examines the distribution of Sex, ChestPainType, and FastingBS in relation to the HeartDisease target. A correlation heatmap for numerical variables is also generated to inspect relationships among the numerical features.

## 7. Machine Learning Models

Five classification algorithms were trained and evaluated using the prepared training and testing data:

| Model | Accuracy | F1 Score |
|---|---|---|
| Logistic Regression | 86.96% | 88.57% |
| K-Nearest Neighbors (KNN) | 86.41% | 88.15% |
| Naive Bayes | 85.33% | 86.83% |
| Support Vector Machine (SVM) | 84.78% | 86.79% |
| Decision Tree | 76.63% | 79.23% |

## 8. Results and Discussion

The results recorded in the notebook show that Logistic Regression achieved the highest accuracy among the five evaluated models, with an accuracy of 0.8696 (86.96%) and an F1-score of 0.8857 (88.57%). KNN was the second-highest model by accuracy at 86.41%. Naive Bayes achieved 85.33%, while SVM achieved 84.78%. Decision Tree recorded the lowest accuracy at 76.63%. Based on these recorded results, Logistic Regression is the best-performing model among the models tested in this project.

## 9. Conclusion

The project demonstrates a complete basic machine learning workflow for a binary classification problem: dataset inspection, cleaning, exploratory analysis, categorical encoding, scaling, train-test splitting, model training, and performance comparison. Five classification algorithms were tested, with Logistic Regression giving the best recorded performance at 86.96% accuracy and an F1-score of 88.57%. The results show that machine learning models can be applied to the available patient features to build a heart disease prediction classifier.

## 10. Future Scope

Future work can include testing additional classification algorithms, performing systematic hyperparameter tuning, applying cross-validation, studying feature importance, and evaluating the models with additional metrics such as precision, recall, ROC-AUC, and confusion matrices. A more carefully validated preprocessing pipeline could also be used to make the evaluation more robust.

## 11. Tools and Technologies

**Programming Language:** Python
**Environment:** Jupyter Notebook
**Libraries:** Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn
**Dataset File:** heart.csv
**Notebook File:** heart.ipynb

## 12. Project Files

The GitHub repository for this project can contain the following files: heart.ipynb (main notebook), heart.csv (dataset), requirements.txt (Python dependencies), README.md (project documentation), and this report as report.pdf.

## 13. References

1. Project dataset: heart.csv supplied with the project.
2. Project implementation and results: heart.ipynb supplied with the project.
3. Scikit-learn library documentation for the machine learning algorithms used.